Chapter 4: Transmission Lines and Signal Integrity

Source: chap4.pdf
Topics: Transmission-line equations, per-unit-length parameters, time-domain solution, signal integrity, sinusoidal/phasor analysis, lossy lines, lumped-circuit models

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1. Preview

Why Transmission Lines Matter

(chap4.pdf, p.3)

- A transmission line imposes a **time delay** as signals propagate from one end to the other:

$$T_D = \frac{\mathscr{L}}{v}$$

(chap4.pdf, p.3)

where $\mathscr{L}$ is the line length and v is the velocity of propagation.

- **Impedance mismatch** causes reflections, which distort the original signal — the primary mechanism degrading signal integrity.
- **Signal integrity** means the waveforms at the input and output of a line are identical or approximately so.

Categories of Transmission Lines

(chap4.pdf, p.4–5)

Wire-type transmission lines:

Structure	Dispersive?	Notes
Two-wire	Yes	Dispersive due to outer insulator
Wire above ground plane	Yes	Dispersive due to outer insulator
Coaxial cable	No	Not dispersive

Printed Circuit Board (PCB) structures:

Structure	Dispersive?	Notes
Stripline (inner-plane boards)	No	Field fully enclosed in dielectric
Microstrip (outer lands of inner-plane board)	Yes	Inhomogeneous medium
Coplanar strip (lands on board without inner planes)	Yes	Inhomogeneous medium

Figure descriptions: (p.4) Three wire-type TX line cross sections shown as (a) two-wire, (b) wire above ground, (c) coaxial. (p.5) Three PCB structures: (a) stripline between two ground planes, (b) microstrip on substrate above ground, (c) coplanar strips on substrate.

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2. The Transmission-Line Equations

Lossless Equivalent Circuit Model

(chap4.pdf, p.6–7)

A two-conductor transmission line is modeled as a **distributed parameter circuit**: a sequence of series inductors ($l\Delta z$ each) and shunt capacitors ($c\Delta z$ each), where l and c are per-unit-length inductance (H/m) and capacitance (F/m).

Figure (p.6): Distributed circuit model showing repeated $l\Delta z$ inductors in series and $c\Delta z$ capacitors to ground for each Δz cell.

Physical interpretation: - **Effect of L and C:** Energizing/de-energizing inductors and capacitors takes finite time $\rightarrow$ signal propagates at finite speed $\rightarrow$ time delay $T_D = \mathscr{L}/v$. - **Effect of R:** At GHz frequencies, conductor resistance becomes significant due to skin effect and cannot be neglected.

Telegrapher's Equations — First-Order (Coupled) Form

(chap4.pdf, p.8–9)

Applying **KVL** around the outer loop of a Δz section and taking $\lim_{\Delta z \to 0}$:

$$\frac{\partial V(z,t)}{\partial z} = -l \frac{\partial I(z,t)}{\partial t} \tag{1}$$

(chap4.pdf, p.8)

Applying **KCL** at the upper node of the capacitor:

$$\frac{\partial I(z,t)}{\partial z} = -c \frac{\partial V(z,t)}{\partial t} \tag{2}$$

(chap4.pdf, p.9)

Equations (1) and (2) are the **coupled transmission-line equations** in first-order form (Telegrapher's equations).

Telegrapher's Equations — Second-Order (Uncoupled) Form

(chap4.pdf, p.10)

Differentiating (1) with respect to z and (2) with respect to t, then combining:

$$\frac{\partial^2 V(z,t)}{\partial z^2} = lc \frac{\partial^2 V(z,t)}{\partial t^2}$$

(chap4.pdf, p.10)

$$\frac{\partial^2 I(z,t)}{\partial z^2} = lc \frac{\partial^2 I(z,t)}{\partial t^2}$$

(chap4.pdf, p.10)

These are the **uncoupled transmission-line equations** in second-order form (wave equations).

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3. The Per-Unit-Length Parameters

TEM Wave and Field Derivation

(chap4.pdf, p.11)

Transmission lines support **Transverse ElectroMagnetic (TEM)** waves: no field components in the direction of propagation ($H_z = 0$, $E_z = 0$).

- From Faraday's law with $H_z = 0$: derive transverse electric field $\vec{E}_T$
- From Ampere's law with $E_z = 0$: derive transverse magnetic field $\vec{H}_T$

- From $\vec{E}_T$ and $\vec{H}_T$, compute per-unit-length l and c using **dc field computations** (valid because TEM fields have the same spatial distribution as static fields)

Homogeneous Medium Relations

(chap4.pdf, p.12)

For a **homogeneous** surrounding medium:

$$lc = \mu\epsilon$$

(chap4.pdf, p.12)

This yields:

$$l = \frac{1}{cv^2}, \quad c = \frac{1}{lv^2}$$

where the velocity of propagation is:

$$v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{v_0}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{\sqrt{\epsilon_r}} \text{ m/s}$$

(chap4.pdf, p.12)

Effective Relative Permittivity (Inhomogeneous Medium)

(chap4.pdf, p.13)

For inhomogeneous media (e.g., microstrip with air above and dielectric below), replace ϵ_r with an **effective relative permittivity** ϵ_r':

$$lc = \mu_0\epsilon_0\epsilon_r'$$

$$v = \frac{1}{\sqrt{\mu_0\epsilon_0\epsilon_r'}} = \frac{v_0}{\sqrt{\epsilon_r'}} \text{ m/s}$$

(chap4.pdf, p.13)

Wire-Type Structure Formulas

Magnetic Field of a Current-Carrying Wire *(chap4.pdf, p.14)*

By Ampere's law (uniformly distributed current):

$$H_T = \frac{I}{2\pi r} \text{ (A/m)}$$

(chap4.pdf, p.14)

Magnetic Flux (Used to Find l) *(chap4.pdf, p.15)*

$$\psi_m = \int_S \vec{B}_T \cdot d\vec{s} = \int_{R_1}^{R_2} \frac{\mu_0 I}{2\pi r} dr = \frac{\mu_0 I}{2\pi} \ln\left(\frac{R_2}{R_1}\right) \text{ (Wb)}$$

(chap4.pdf, p.15)

Per-unit-length inductance: $\Psi = lI$, so $l = \psi_m / I$.

Electric Field of a Charged Wire *(chap4.pdf, p.16)*

By Gauss's law:

$$E_T = \frac{q}{2\pi\epsilon_0 r} \text{ (V/m)}$$

(chap4.pdf, p.16)

Voltage Between Two Points (Used to Find c) *(chap4.pdf, p.17)*

$$V = \frac{q}{2\pi\epsilon_0} \ln\left(\frac{R_2}{R_1}\right) \text{ (V)}, \quad R_2 \geq R_1$$

(chap4.pdf, p.17)

Per-unit-length capacitance: $q = cV$, so $c = q/V$.

Two-Wire Line

(chap4.pdf, p.18–22)

For two wires of radius r_w separated by distance s ($s \gg r_w$):

Approximate per-unit-length inductance:

$$l \approx \frac{\mu_0}{\pi} \ln\left(\frac{s}{r_w}\right) = 0.4 \ln\left(\frac{s}{r_w}\right) \text{ (μH/m)} = 10.16 \ln\left(\frac{s}{r_w}\right) \text{ (nH/in.)}$$

(chap4.pdf, p.19)

Exact per-unit-length inductance:

$$l = \frac{\mu_0}{\pi} \cosh^{-1}\left(\frac{s}{2r_w}\right) = \frac{\mu_0}{\pi} \ln\left[\frac{s}{2r_w} + \sqrt{\left(\frac{s}{2r_w}\right)^2 - 1}\right] \text{ (H/m)}$$

(chap4.pdf, p.20)

Approximate per-unit-length capacitance:

$$c \approx \frac{\pi\epsilon_0}{\ln(s/r_w)} = \frac{27.78}{\ln(s/r_w)} \text{ (pF/m)} = \frac{0.706}{\ln(s/r_w)} \text{ (pF/in.)}$$

(chap4.pdf, p.22)

Exact per-unit-length capacitance:

$$c = \frac{\pi\epsilon_0}{\cosh^{-1}(s/2r_w)} \text{ (F/m)}$$

(chap4.pdf, p.20)

Note: $\cosh^{-1}(x) \approx \ln(2x)$ for $x \gg 1$.

Wire Above a Ground Plane

(chap4.pdf, p.23–24)

Using the **method of images**: capacitance between wire (height h, radius r_w) and ground plane equals twice the capacitance of two wires at separation $2h$.

Exact per-unit-length capacitance:

$$c = \frac{2\pi\epsilon_0}{\cosh^{-1}(h/r_w)} \approx \frac{2\pi\epsilon_0}{\ln(2h/r_w)} \text{ (F/m)}, \quad h \gg r_w$$

(chap4.pdf, p.24)

Exact per-unit-length inductance (half the inductance of the two-wire line):

$$l = \frac{\mu_0}{2\pi}\cosh^{-1}\left(\frac{h}{r_w}\right) \approx \frac{\mu_0}{2\pi}\ln\left(\frac{2h}{r_w}\right) \text{ (H/m)}, \quad h \gg r_w$$

(chap4.pdf, p.24)

Coaxial Cable

(chap4.pdf, p.25–27)

For inner wire of radius r_w and outer shield of inner radius r_s, no proximity effect due to symmetry.

Exact per-unit-length inductance:

$$l = \frac{\mu_0}{2\pi}\ln\left(\frac{r_s}{r_w}\right) = 0.2\ln\left(\frac{r_s}{r_w}\right) \text{ (μH/m)} = 5.08\ln\left(\frac{r_s}{r_w}\right) \text{ (nH/in.)}$$

(chap4.pdf, p.26)

Exact per-unit-length capacitance:

$$c = \frac{2\pi\epsilon}{\ln(r_s/r_w)} = \frac{55.56\epsilon_r}{\ln(r_s/r_w)} \text{ (pF/m)} = \frac{1.4\epsilon_r}{\ln(r_s/r_w)} \text{ (pF/in.)}$$

(chap4.pdf, p.27)

PCB Structures — Characteristic Impedance

(chap4.pdf, p.28)

For PCB lines, work with **characteristic impedance** Z_C and **velocity** v:

$$Z_C = \sqrt{\frac{l}{c}}\ \Omega, \quad v = \frac{1}{\sqrt{lc}} = \frac{v_0}{\sqrt{\epsilon_r'}} \text{ m/s}$$

(chap4.pdf, p.28)

From which: $l = Z_C/v$ and $c = 1/(vZ_C)$.

Stripline (inner-plane boards, $t = 0$) *(chap4.pdf, p.29)*

$$Z_C = \frac{30\pi}{\sqrt{\epsilon_r}} \cdot \frac{1}{\left[\frac{w_e}{s} + 0.441\right]}$$

(chap4.pdf, p.29)

where s is board thickness between planes, w is land width, and

$$\frac{w_e}{s} = \begin{cases} \frac{w}{s} & \frac{w}{s} \geq 0.35 \\ \frac{w}{s} - \left(0.35 - \frac{w}{s}\right)^2 & \frac{w}{s} \leq 0.35 \end{cases}$$

Note: $\epsilon_r' = \epsilon_r$ for stripline (homogeneous medium).

Microstrip Line (outer lands of inner-plane board, $t = 0$) *(chap4.pdf, p.30)*

$$Z_C = \begin{cases} \frac{60}{\sqrt{\epsilon_r'}} \ln\left[\frac{8h}{w} + \frac{w}{4h}\right] & \frac{w}{h} \leq 1 \\ \frac{120\pi}{\sqrt{\epsilon_r'}} \left[\frac{w}{h} + 1.393 + 0.667 \ln\left(\frac{w}{h} + 1.444\right)\right]^{-1} & \frac{w}{h} \geq 1 \end{cases}$$

(chap4.pdf, p.30)

where h is substrate thickness, w is land width, and the **effective relative permittivity** is:

$$\epsilon_r' = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \cdot \frac{1}{\sqrt{1 + 10h/w}}$$

(chap4.pdf, p.30)

Limiting cases: $\epsilon_r' \approx \epsilon_r$ for $h \ll w$; $\epsilon_r' \approx (\epsilon_r + 1)/2$ for $h \gg w$.

Coplanar Strip (lands on board without inner planes, $t = 0$) *(chap4.pdf, p.31)*

$$Z_C = \begin{cases} \dfrac{120}{\sqrt{\epsilon_r'}} \ln\left(2\dfrac{1+\sqrt{k}}{1-\sqrt{k}}\right) & \dfrac{1}{\sqrt{2}} \le k \le 1 \\ \dfrac{377\pi}{\sqrt{\epsilon_r'} \ln\left(2\dfrac{1+\sqrt{k'}}{1-\sqrt{k'}}\right)} & 0 \le k \le \dfrac{1}{\sqrt{2}} \end{cases}$$

(chap4.pdf, p.31)

where $k = s/(s + 2w)$ and $k' = \sqrt{1 - k^2}$.

Strips on Opposite Sides (power plane configuration) *(chap4.pdf, p.32)*

Two formulas depending on $w/h > 1$ or $w/h < 1$. Wide lands on opposite sides give **lower characteristic impedance** than on the same side — useful for power planes where low inductance ($v = L\, di/dt$) is needed.

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4. The Time-Domain Solution

Definition

(chap4.pdf, p.33)

The time-domain solution is the **complete solution** (transient + steady state) with no assumptions about the time form of excitation. It is the primary tool for signal integrity analysis.

Lossless Line Solution

(chap4.pdf, p.34)

The general solution of the wave equations consists of forward- and backward-traveling waves:

$$V(z,t) = V^+\left(t - \frac{z}{v}\right) + V^-\left(t + \frac{z}{v}\right)$$

(chap4.pdf, p.34)

$$I(z,t) = \frac{1}{Z_C} V^+\left(t - \frac{z}{v}\right) - \frac{1}{Z_C} V^-\left(t + \frac{z}{v}\right)$$

(chap4.pdf, p.34)

where: - V^+: forward-traveling wave (in $+z$ direction) - V^-: backward-traveling wave (in $-z$ direction) - Characteristic impedance: $Z_C = \sqrt{l/c} = vl = 1/(vc)$ - Velocity: $v = 1/\sqrt{lc} = 1/\sqrt{\mu\epsilon}$

Each wave's current is related to its voltage by Z_C:

$$I^+\left(t - \frac{z}{v}\right) = \frac{1}{Z_C} V^+\left(t - \frac{z}{v}\right)$$

$$I^-\left(t + \frac{z}{v}\right) = -\frac{1}{Z_C} V^-\left(t + \frac{z}{v}\right)$$

(chap4.pdf, p.35)

Load and Source Reflection Coefficients

(chap4.pdf, p.35–37)

Load reflection coefficient (at $z = \mathscr{L}$):

$$\Gamma_L = \frac{V^-(t + \mathscr{L}/v)}{V^+(t - \mathscr{L}/v)} = \frac{R_L - Z_C}{R_L + Z_C}$$

(chap4.pdf, p.35)

Source reflection coefficient (at $z = 0$):

$$\Gamma_S = \frac{R_S - Z_C}{R_S + Z_C}$$

(chap4.pdf, p.37)

The current reflection coefficient is the **negative** of the voltage reflection coefficient.

Initially launched voltage and current from source:

$$V(0,t) = \frac{Z_C}{R_S + Z_C} V_S(t), \quad I(0,t) = \frac{V_S(t)}{R_S + Z_C}$$

(chap4.pdf, p.37)

The backward-traveling wave is related to the forward wave at the load:

$$V^-\left(t + \frac{\mathscr{L}}{v}\right) = \Gamma_L V^+\left(t - \frac{\mathscr{L}}{v}\right)$$

(chap4.pdf, p.36)

Multiple reflections continue bouncing between source and load.

Matched case ($R_L = Z_C$): Only a time delay is imposed; the line "doesn't matter."

$$V(0,t) = \frac{Z_C}{R_S + Z_C} V_S(t), \quad V(\mathscr{L},t) = \frac{Z_C}{R_S + Z_C} V_S(t - T_D)$$

(chap4.pdf, p.43)

Bounce (Lattice) Diagram

(chap4.pdf, p.42–43)

The bounce diagram is a graphical tool for tracking multiple reflections: - Horizontal axis: position z from 0 to $\mathscr{L}$ - Vertical axis: time - Diagonal lines represent forward/backward wave bounces - At each bounce: multiply by Γ_L (at load) or Γ_S (at source)

Voltage at $z = 0$ (source end):

$$V(0,t) = \frac{Z_C}{R_S + Z_C} \left[V_S(t) + (1+\Gamma_S)\Gamma_L V_S(t-2T_D) + (1+\Gamma_S)(\Gamma_S\Gamma_L)\Gamma_L V_S(t-4T_D) + \cdots \right]$$

(chap4.pdf, p.43)

Voltage at $z = \mathscr{L}$ (load end):

$$V(\mathscr{L},t) = \frac{Z_C}{R_S + Z_C} \left[(1+\Gamma_L) V_S(t - T_D) + (1+\Gamma_L)\Gamma_S\Gamma_L V_S(t - 3T_D) + \cdots \right]$$

(chap4.pdf, p.43)

Example 4.1: Voltage on a Line as a Function of t

(chap4.pdf, p.38)

Circuit: $V_S = 30$ V, $R_S = 0$ ($\Gamma_S = -1$), $Z_C = 50\,\Omega$, $v = 200$ m/μs, $\mathscr{L} = 400$ m, $R_L = 100\,\Omega$ ($\Gamma_L = 1/3$), $T_D = 2\,\mu$s.

As time progresses, the load voltage converges to the steady-state value of **30 V**.

Example 4.3: Short Pulsewidth

(chap4.pdf, p.40)

With $T_D = 1\,\mu$s, $R_S = 300\,\Omega$, $Z_C = 100\,\Omega$, $R_L = \infty$:

Initially sent voltage: $\frac{100}{300+100} \times 20 = 5$ V

Both $V(0,t)$ and $V(\mathscr{L},t)$ decay to zero in steady state (as expected for a pulse with no DC component).

Example 4.4: Long Pulsewidth / Ringing

(chap4.pdf, p.41)

Circuit with $R_S = 150\,\Omega$, $Z_C = 50\,\Omega$, $R_L = 0\,\Omega$ ($\Gamma_L = -1$), $\Gamma_S = 1/2$, $T_D = 2\,\mu$s.

Initial voltage: $\frac{50}{150+50} \times 100 = 25$ V

The **overlap of forward and backward waves creates ringing** — a complicated waveform with oscillations.

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5. High-Speed Digital Interconnects and Signal Integrity

Clock Skew

(chap4.pdf, p.44)

The time delay $T_D = \mathscr{L}/v$ in traversing PCB traces becomes critical when it is on the order of the pulse rise/fall times. Unequal path lengths to different clocked elements cause **clock skew** (different arrival times). Careful routing equalizes path lengths.

Impedance Mismatch Effects

(chap4.pdf, p.45–46)

- Mismatched source and load cause reflections → distorted signal
- Changes in trace cross-section change Z_C → reflections at discontinuities
- Layer changes via vias → impedance discontinuity

Typical CMOS gate parameters: - Output resistance: 10–30 Ω - Input capacitance: 5–15 pF

A mismatched line causes ringing in $V(L,t)$, which is caused by multiple reflections.

Effect of Resistive Termination

(chap4.pdf, p.47–49)

Load voltage with resistive termination:

$$V(\mathscr{L},t) = \frac{Z_C}{R_S + Z_C}(1+\Gamma_L)\big[V_S(t-T_D)+\Gamma_S\Gamma_L V_S(t-3T_D)+(\Gamma_S\Gamma_L)^2 V_S(t-5T_D)+\cdots\big]$$

(chap4.pdf, p.48)

Effects of sign combinations of Γ_S and Γ_L:

Γ_S	Γ_L	Condition	Effect
$-$	$+$	$R_S < Z_C$, $R_L > Z_C$	May cause **multiple activations** (overshoot above logic 1)
$+$	$+$	$R_S > Z_C$, $R_L > Z_C$	
$+$	$-$	$R_S > Z_C$, $R_L < Z_C$	May **postpone activation** (undershoot)
$-$	$-$	$R_S < Z_C$, $R_L < Z_C$	

(chap4.pdf, p.49)

Effect of Capacitive Termination

(chap4.pdf, p.50–52)

For a line with matched source ($\Gamma_S = 0$) terminated in capacitor C:

Load reflection coefficient (in Laplace domain):

$$\Gamma_L(s) = \frac{Z_L - Z_C}{Z_L + Z_C} = \frac{1 - sT_C}{1 + sT_C}$$

where $T_C = Z_C C$. *(chap4.pdf, p.50)*

The load voltage (time domain, for step input $V_S = V_0 u(t)$):

$$V_L(t) = V_0 u(t - T_D) - e^{-(t-T_D)/T_C} V_0 u(t - T_D)$$

(chap4.pdf, p.52)

Behavior: - **Initially**: capacitor looks like a **short circuit** - **Steady state**: capacitor transitions to an **open circuit** - **Additional time delay** at the 50% point:

$$t_d = 0.693 T_C = 0.693 C Z_C$$

(chap4.pdf, p.52)

Effect of Inductive Termination

(chap4.pdf, p.53)

For matched source terminated in inductor L:

$$\Gamma_L(s) = \frac{sT_L - 1}{sT_L + 1}, \quad T_L = \frac{L}{Z_C}$$

Load voltage: $V_L(t) = V_0 e^{-(t-T_D)/T_L} u(t - T_D)$

(chap4.pdf, p.53)

Behavior: - **Initially**: inductor looks like an **open circuit** - **Steady state**: inductor transitions to a **short circuit**

Matching Schemes for Signal Integrity

(chap4.pdf, p.54–58)

Series Match *(chap4.pdf, p.54)*

Add resistor R in **series with the source** such that:

$$R_S + R = Z_C$$

(chap4.pdf, p.54)

Advantages: 1. Output reaches same level as input (no attenuation) 2. No DC current flows in the line → **resistor dissipates no power** in steady state

Figure (p.54): Source, series resistor R, transmission line to open-circuit load. Load voltage rises cleanly to V_0 at $t = T_D$, holds at $V_0/2$ until $t = 2T_D$, then rises to V_0.

Parallel Match *(chap4.pdf, p.55)*

Add resistor R in **parallel with the load** such that:

$$R \| R_L = \frac{RR_L}{R + R_L} = Z_C$$

(chap4.pdf, p.55)

Disadvantages: 1. Load voltage is always less than source level V_0 2. Matching resistor **consumes power** even at steady state

Comparison Example *(chap4.pdf, p.56–58)*

Circuit: $R_S = 20\,\Omega$, $Z_C = 50\,\Omega$, $T_D = 0.2$ ns, $C_L = 5$ pF.

- **No match:** Load shows severe ringing (overshoot to ~8 V, undershoot to -1 V for 5 V input)
- **Series match ($R = 30\,\Omega$):** Load rises smoothly to 5 V; three delays sum to:
  - 0.05 ns (source rise time $\tau_r/2$)
  - 0.2 ns (T_D)
  - 0.1733 ns ($0.693CZ_C$)
  - Total = 0.4233 ns
- **Parallel match ($R = 50\,\Omega$):** Still has ringing; capacitor dominates at high frequencies; steady-state voltage falls below 5 V

(chap4.pdf, p.57)

When Does the Line Not Matter?

(chap4.pdf, p.59–63)

Criterion: When the line is electrically short at the highest significant frequency:

$$\mathscr{L} < \frac{1}{10}\left(\frac{v}{f_{\max}}\right)\lambda \quad \Rightarrow \quad T_D = \frac{\mathscr{L}}{v} < \frac{1}{10}\tau_r$$

$$\boxed{\tau_r > 10T_D}$$

(chap4.pdf, p.59)

where $f_{\max} = 1/\tau_r$ is the maximum significant frequency (bandwidth = $1/\tau_r$).

Examples ($R_S = 20\,\Omega$, $Z_C = 50\,\Omega$, $T_D = 0.2$ ns, $C_L = 5$ pF):

τ_r	τ_r/T_D	Overshoot
0.2 ns	$= T_D$	7.0 V (very severe)
1 ns	$= 5T_D$	6.0 V (still significant)
2 ns	$= 10T_D$	5.3 V (moderate)
4 ns	$= 20T_D$	5.2 V (mild)

(chap4.pdf, p.60–63)

As τ_r increases, $f_{\max}$ decreases $\rightarrow$ high-frequency components are reduced $\rightarrow$ overshoot decreases.

Effects of Line Discontinuities

(chap4.pdf, p.64–65)

At a junction between two lines with characteristic impedances Z_{C1} and Z_{C2}:

Reflection/transmission from left (Z_{C1} incident):

$$\Gamma_{12} = \frac{Z_{C2} - Z_{C1}}{Z_{C2} + Z_{C1}}, \quad T_{12} = \frac{2Z_{C2}}{Z_{C1} + Z_{C2}}$$

(chap4.pdf, p.64)

Reflection/transmission from right (Z_{C2} incident):

$$\Gamma_{21} = \frac{Z_{C1} - Z_{C2}}{Z_{C1} + Z_{C2}}, \quad T_{21} = \frac{2Z_{C1}}{Z_{C1} + Z_{C2}}$$

(chap4.pdf, p.65)

Continuity condition: $v_{i1} + v_{r1} = v_{t2}$, so $1 + \Gamma_{12} = T_{12}$.

Example 4.9 (p.67): To eliminate reflections on lines with discontinuities, must use **series match at the source AND parallel match at the load** simultaneously.

Effects of Feedline Discontinuities

(chap4.pdf, p.68–72)

Series multiple feedlines (daisy chain): - Series match at source ($R_S + R = Z_C$): eliminates all reflections - Parallel match at midpoint: does NOT eliminate reflections (because $\Gamma_{12} = -1/3$, not zero)

Parallel multiple feedlines (bus topology): - For N equal branches each with impedance Z_C: the parallel combination is Z_C/N - Series match at source: $R_S = Z_C/N$ — works only if all time delays are equal ($T_{D1} = T_{D2}$) - Parallel match at loads (Z_C across each load): works but dissipates power

(chap4.pdf, p.72)

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6. Sinusoidal Excitation and the Phasor Solution

Phasor Form of Lossless TX Line

(chap4.pdf, p.73)

Starting from Telegrapher's equations and converting to phasor/frequency domain ($\partial/\partial t \to j\omega$):

$$\frac{d\hat{V}(z)}{dz} = -j\omega l\hat{I}(z), \quad \frac{d\hat{I}(z)}{dz} = -j\omega c\hat{V}(z)$$

(chap4.pdf, p.73)

Second-order uncoupled equations:

$$\frac{d^2\hat{V}(z)}{dz^2} + \omega^2 lc\,\hat{V}(z) = 0, \quad \frac{d^2\hat{I}(z)}{dz^2} + \omega^2 lc\,\hat{I}(z) = 0$$

Phasor solution (forward + backward waves):

$$\hat{V}(z) = \hat{V}^+ e^{-j\beta z} + \hat{V}^- e^{j\beta z}$$

$$\hat{I}(z) = \frac{\hat{V}^+}{Z_C} e^{-j\beta z} - \frac{\hat{V}^-}{Z_C} e^{j\beta z}$$

(chap4.pdf, p.73)

Characteristic Impedance and Phase Constant

(chap4.pdf, p.74)

$$Z_C = \sqrt{\frac{l}{c}} = vl = \frac{1}{vc}, \quad \beta = \frac{\omega}{v} = \omega\sqrt{lc} = \frac{2\pi}{\lambda}$$

(chap4.pdf, p.74)

Voltage Reflection Coefficient

(chap4.pdf, p.74)

The voltage reflection coefficient at position z:

$$\hat{\Gamma}(z) = \frac{\hat{V}^- e^{j\beta z}}{\hat{V}^+ e^{-j\beta z}} = \frac{\hat{V}^-}{\hat{V}^+} e^{j2\beta z}$$

At the load ($z = \mathscr{L}$):

$$\hat{\Gamma}_L = \frac{\hat{Z}_L - Z_C}{\hat{Z}_L + Z_C}$$

(chap4.pdf, p.74)

Reflection coefficient at arbitrary point z in terms of load reflection:

$$\hat{\Gamma}(z) = \hat{\Gamma}_L e^{j2\beta(z-\mathscr{L})}$$

(chap4.pdf, p.74)

Voltage and Current in Terms of $\hat{\Gamma}$

(chap4.pdf, p.75)

$$\hat{V}(z) = \hat{V}^+ e^{-j\beta z}\big[1 + \hat{\Gamma}(z)\big]$$

$$\hat{I}(z) = \frac{\hat{V}^+}{Z_C} e^{-j\beta z}\big[1 - \hat{\Gamma}(z)\big]$$

Input Impedance

(chap4.pdf, p.75–76)

$$\hat{Z}_{\text{in}}(z) = \frac{\hat{V}(z)}{\hat{I}(z)} = Z_C \frac{1 + \hat{\Gamma}(z)}{1 - \hat{\Gamma}(z)}$$

(chap4.pdf, p.75)

At the source end ($z = 0$), with $\hat{\Gamma}(0) = \hat{\Gamma}_L e^{-j2\beta\mathscr{L}}$:

$$\hat{Z}_{\text{in}}(0) = Z_C \frac{1 + \hat{\Gamma}_L e^{-j2\beta\mathscr{L}}}{1 - \hat{\Gamma}_L e^{-j2\beta\mathscr{L}}} = Z_C \frac{\hat{Z}_L + jZ_C \tan\beta\mathscr{L}}{Z_C + j\hat{Z}_L \tan\beta\mathscr{L}}$$

(chap4.pdf, p.76)

Voltage Standing-Wave Ratio (VSWR)

(chap4.pdf, p.77)

$$\text{VSWR} = \frac{|\hat{V}(z)|_{\max}}{|\hat{V}(z)|_{\min}} = \frac{1 + |\hat{\Gamma}_L|}{1 - |\hat{\Gamma}_L|}, \quad 1 \le \text{VSWR} \le \infty$$

(chap4.pdf, p.77)

- VSWR = 1: matched line ($\Gamma_L = 0$), forward wave only
- VSWR = ∞: total reflection ($|\Gamma_L| = 1$, open or short circuit load)

When matched ($\hat{Z}_L = Z_C$, $\Gamma_L = 0$):

$$\hat{V}(z) = \frac{Z_C}{Z_C + \hat{Z}_S} \hat{V}_S e^{-j\beta z}, \quad \hat{I}(z) = \frac{1}{Z_C + \hat{Z}_S} \hat{V}_S e^{-j\beta z}$$

(chap4.pdf, p.77)

Power Flow

(chap4.pdf, p.78)

Average power flow in the $+z$ direction:

$$P_{\text{av}}(z) = \frac{1}{2}\mathscr{R}\{\hat{V}(z)\hat{I}^*(z)\} = \frac{1}{2}\frac{|\hat{V}^+|^2}{Z_C}\left(1 - |\hat{\Gamma}_L|^2\right)$$

(chap4.pdf, p.78)

Ratio of reflected to incident power:

$$\frac{P_{\text{av, reflected}}}{P_{\text{av, incident}}} = |\hat{\Gamma}_L|^2$$

(chap4.pdf, p.78)

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7. Phasor Form of Lossy TX Line

Lossy Telegrapher's Equations

(chap4.pdf, p.79–80)

For a **lossy** TX line, the equivalent circuit adds series resistance r (Ω/m) and shunt conductance g (S/m):

$$\frac{d\hat{V}(z)}{dz} = -\hat{z}\hat{I}(z), \quad \frac{d\hat{I}(z)}{dz} = -\hat{y}\hat{V}(z)$$

(chap4.pdf, p.80)

where the per-unit-length impedance and admittance are:

$$\hat{z} = r(f) + j\omega l, \quad \hat{y} = g + j\omega c$$

(chap4.pdf, p.80)

Note: $r(f)$ is frequency-dependent due to **skin effect**.

Solution of Lossy Line

(chap4.pdf, p.80–81)

$$\hat{V}(z) = \hat{V}^+ e^{-\alpha z} e^{-j\beta z} + \hat{V}^- e^{\alpha z} e^{j\beta z}$$

$$\hat{I}(z) = \frac{\hat{V}^+}{\hat{Z}_C} e^{-\alpha z} e^{-j\beta z} - \frac{\hat{V}^-}{\hat{Z}_C} e^{\alpha z} e^{j\beta z}$$

(chap4.pdf, p.80)

Propagation Constant and Characteristic Impedance

(chap4.pdf, p.81)

Propagation constant:

$$\hat{\gamma} = \sqrt{\hat{z}\hat{y}} = \sqrt{(r + j\omega l)(g + j\omega c)} = \alpha + j\beta$$

(chap4.pdf, p.81)

- α: attenuation constant (Np/m)
- β: phase constant (rad/m)

Characteristic impedance:

$$\hat{Z}_C = \sqrt{\frac{\hat{z}}{\hat{y}}} = \sqrt{\frac{r(f) + j\omega l}{g + j\omega c}}$$

(chap4.pdf, p.81)

Forward and backward waves are **attenuated** as they travel along the line.

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8. Effect of Losses on Signal Integrity

Skin Effect — Circular Wire

(chap4.pdf, p.82–83)

DC resistance per unit length:

$$r_{\text{dc}} = \frac{1}{\sigma \pi r_w^2} \ \Omega/\text{m}$$

(chap4.pdf, p.82)

As frequency increases, current concentrates in an annulus of thickness equal to the **skin depth**:

$$\delta = \frac{1}{\sqrt{\pi f \mu_0 \sigma}} \ \text{m}$$

(chap4.pdf, p.82)

Skin depth **decreases** as $\sqrt{f}$.

For $r_w \gg \delta$ (high frequency):

$$r_{\text{hf}} = \frac{1}{\sigma(\pi r_w^2 - \pi(r_w - \delta)^2)} \approx \frac{1}{\sigma 2\pi r_w \delta}\ \Omega/\text{m}$$

(chap4.pdf, p.83)

$r(f)$ **increases as** $\sqrt{f}$ (or 10 dB/decade). The transition frequency from r_{dc} to r_{hf} is determined by setting $r_{\text{dc}} = r_{\text{hf}}$.

Figure (p.83): Graph of $r(\Omega/m)$ vs. f showing flat r_{dc} region then rising at 10 dB/decade after f_{break}.

Skin Effect — Rectangular PCB Land

(chap4.pdf, p.84)

For a rectangular land of width w and thickness t:

DC resistance:

$$r_{\text{dc}} = \frac{1}{\sigma w t}\ \Omega/\text{m}$$

(chap4.pdf, p.84)

At high frequency (current in annulus of depth δ):

$$r_{\text{hf}} = \frac{1}{\sigma(2\delta t + 2\delta w)} = \frac{1}{2\sigma\delta(w+t)}\ \Omega/\text{m}$$

(chap4.pdf, p.84)

Transition skin depth: $\delta_{\text{break}} = \frac{1}{2}\frac{wt}{w+t} \approx \frac{t}{2}$ for $w \gg t$.

Current also peaks at the **corners** of the rectangular land.

Dielectric Loss

(chap4.pdf, p.85–87)

Dielectrics have **bound charge** (microscopic dipoles) but no appreciable free charge. Dielectric loss arises from inability of dipoles to follow time-varying electric fields.

Complex permittivity:

$$\epsilon_r = \epsilon_r' - j\epsilon_r''$$

(chap4.pdf, p.86)

The per-unit-length admittance becomes:

$$j\omega c = j\omega\epsilon_0\epsilon_r K = \underbrace{\omega\epsilon_0\epsilon_r'' K}_{g} + j\,\underbrace{\omega\epsilon_0\epsilon_r' K}_{c}$$

(chap4.pdf, p.86)

Loss tangent:

$$\tan\theta = \frac{\epsilon_r''}{\epsilon_r'} \approx \frac{g}{\omega c} \quad \Rightarrow \quad g = \omega c \tan\theta$$

(chap4.pdf, p.87)

For FR-4: $\tan\theta \approx 0.02$.

Dielectric conductance g increases **directly with frequency** (20 dB/decade), unlike skin-effect resistance (10 dB/decade).

Both α and β are frequency-dependent $\rightarrow$ **two kinds of dispersion**.

Figure (p.94): Graph showing $r(f)$ at 10 dB/decade and $g = \omega c \tan\theta$ at 20 dB/decade.

Signal Distortion Due to Losses

(chap4.pdf, p.88)

- **Phase dispersion** (β frequency-dependent): sinusoidal components travel at different speeds $\rightarrow$ arrive at different times $\rightarrow$ pulse distortion
- **Attenuation dispersion** (α frequency-dependent): components attenuated by different amounts $\rightarrow$ pulse shape distortion
- **General effect**: high-frequency components attenuated more $\rightarrow$ pulse bandwidth reduced $\rightarrow$ rise/fall times increased

Low-Loss Approximation

(chap4.pdf, p.89–91)

Condition for low-loss:

$$r \ll \omega l, \quad g \ll \omega c$$

(chap4.pdf, p.90)

Simplified propagation constant:

$$\hat{\gamma} \approx j\omega\sqrt{lc}\left[1 - j\left(\frac{r}{2\omega l} + \frac{g}{2\omega c}\right)\right]$$

(chap4.pdf, p.90)

Attenuation constant (low-loss):

$$\alpha \approx \frac{1}{2}\left(\frac{r}{Z_C} + gZ_C\right)$$

(chap4.pdf, p.91)

- Constant at low frequency (dominated by r_{dc})

Phase constant (low-loss):

$$\beta \approx \omega\sqrt{lc}$$

(chap4.pdf, p.91)

- Linear function of frequency → no phase dispersion in low-loss regime

Characteristic impedance (low-loss):

$$Z_C \approx \sqrt{\frac{l}{c}} \quad \text{(independent of frequency)}$$

(chap4.pdf, p.91)

Cable Loss Formula

(chap4.pdf, p.89)

For a matched load, power is attenuated as $e^{-2\alpha\mathscr{L}}$, giving:

$$\text{Loss}_{\text{dB}} = 8.686\alpha\mathscr{L} \text{ dB}$$

(chap4.pdf, p.89)

Typical PCB Board Example

(chap4.pdf, p.92–94)

For typical board dimensions: - Low-loss region (both $r \ll \omega l$ and $g \ll \omega c$): above ~**5 MHz** - Region with no α or β dispersion: 5–20 MHz - Above 20 MHz: no β dispersion, but α dispersion present (skin effect)

Figures (p.92–93): Characteristic impedance magnitude decreases from 100s Ω below 5 MHz to ~50 Ω above 5 MHz. Attenuation constant shows transition at 5 MHz and 20 MHz. Velocity of propagation rises and stabilizes above 5 MHz.

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9. Lumped-Circuit Approximate Models

When to Use Lumped Models

(chap4.pdf, p.95)

Use lumped-circuit models when the line is **electrically short** at the highest significant frequency:

$$\mathscr{L} < \frac{\lambda}{10} \text{ at } f_{\max}$$

(chap4.pdf, p.95)

Lumped-π and Lumped-T Models

(chap4.pdf, p.95)

Two standard approximations for a transmission line of length $\mathscr{L}$:

Lumped-π model: Series $r\mathscr{L}$, $l\mathscr{L}$ in center; shunt $g\mathscr{L}/2$, $c\mathscr{L}/2$ on each side.

Lumped-T model: Series $r\mathscr{L}/2$, $l\mathscr{L}/2$ on each end; shunt $g\mathscr{L}$, $c\mathscr{L}$ in center.

Figure (p.95): Circuit diagrams of lumped-π and lumped-T models.

Note: Difficult to use in time-domain since losses are not easily handled.

Example 4.12: Number of Subsections Required

(chap4.pdf, p.96–97)

For a 1 m lossless line ($Z_C = 50\,\Omega$, $v = 2 \times 10^8$ m/s):

- **1-section lumped-π**: valid up to **10 MHz** (line is $\lambda/10$ at 10 MHz)
- **2-section lumped-π**: valid up to **20 MHz** (each section is $\lambda/10$ at 20 MHz)

Each additional section doubles the useful frequency range. Each section should be $\leq \lambda/10$.

Summary Tables

Key Equations Summary

Quantity	Formula	Location
Time delay	$T_D = \mathscr{L}/v$	p.3
Telegrapher's eq. (KVL)	$\partial V/\partial z = -l\,\partial I/\partial t$	p.8
Telegrapher's eq. (KCL)	$\partial I/\partial z = -c\,\partial V/\partial t$	p.9
Wave equation (voltage)	$\partial^2 V/\partial z^2 = lc\,\partial^2 V/\partial t^2$	p.10

Quantity	Formula	Location
$lc = \mu\epsilon$ relation	$v = 1/\sqrt{lc} = v_0/\sqrt{\epsilon_r}$	p.12
Characteristic impedance	$Z_C = \sqrt{l/c}$	p.28, 34
Load reflection coeff.	$\Gamma_L = (R_L - Z_C)/(R_L + Z_C)$	p.35
Source reflection coeff.	$\Gamma_S = (R_S - Z_C)/(R_S + Z_C)$	p.37
Initial launched voltage	$V(0,t) = \frac{Z_C}{R_S+Z_C} V_S(t)$	p.37
Series match condition	$R_S + R = Z_C$	p.54
Parallel match condition	$R \| R_L = Z_C$	p.55
TL negligible criterion	$\tau_r > 10T_D$	p.59
Phase constant	$\beta = \omega/v = \omega\sqrt{lc}$	p.74
VSWR	$(1 + \|\Gamma_L\|)/(1 - \|\Gamma_L\|)$	p.77
Input impedance	$Z_{\text{in}} = Z_C(Z_L + jZ_C \tan\beta\mathscr{L})/(Z_C + jZ_L \tan\beta\mathscr{L})$	p.76
Propagation constant	$\hat{\gamma} = \sqrt{(r + j\omega l)(g + j\omega c)} = \alpha + j\beta$	p.81
Skin depth	$\delta = 1/\sqrt{\pi f \mu_0 \sigma}$	p.82
Cable loss	$8.686\alpha\mathscr{L}$ dB	p.89
Low-loss criterion	$r \ll \omega l$, $g \ll \omega c$	p.90
Loss tangent	$\tan\theta = \epsilon_r''/\epsilon_r'$, $g = \omega c \tan\theta$	p.87

Per-Unit-Length Parameter Summary

Structure	l (H/m)	c (F/m)
Two-wire ($s \gg r_w$)	$\frac{\mu_0}{\pi} \ln(s/r_w)$	$\frac{\pi\epsilon_0}{\ln(s/r_w)}$
Wire above ground ($h \gg r_w$)	$\frac{\mu_0}{2\pi} \ln(2h/r_w)$	$\frac{2\pi\epsilon_0}{\ln(2h/r_w)}$
Coaxial cable	$\frac{\mu_0}{2\pi} \ln(r_s/r_w)$	$\frac{2\pi\epsilon}{\ln(r_s/r_w)}$

Special Load Reflection Coefficients

Load	Γ_L	Behavior
Matched ($Z_L = Z_C$)	0	No reflection
Open circuit ($Z_L = \infty$)	+1	Full reflection, same polarity

Load	Γ_L	Behavior
Short circuit ($Z_L = 0$)	-1	Full reflection, reversed polarity
Inductive ($Z_L = j\omega L$)	$+1$ at $t = 0^+$, 0 at $t \to \infty$ (for matched source)	Initially open, then matched
Capacitive ($Z_L = 1/j\omega C$)	-1 at $t = 0^+$, $+1$ at $t \to \infty$ (for matched source)	Initially short, then open